

TITLE OF THE PROJECT

MINOR PROJECT REPORT

**Submitted in partial fulfillment of the requirement for the Degree of
Bachelors of Engineering in Computer Science & Engineering**

Submitted To:



[PARUL UNIVERSITY, VADODARA, GUJARAT (INDIA)]

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

PARUL INSTITUTE OF TECHNOLOGY VADODARA, GUJARAT

SESSION: AY 2025-2026

Parul University

Parul Institute of Technology



(Session: 2024 -2025)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CERTIFICATE

This is to certify that **Anshuman Shukla, Prakhar Batwal, Auchitya Kumar mishra** , Students of **CSE VI Semester** of “ **Parul Institute of Technology, Vadodara**” has completed their **Minor Project** titled “**TASKFLOW**”, as per the syllabus and has submitted a satisfactory report on this project as a partial fulfillment towards the award of degree of **Bachelor of Technology in Computer Science and Engineering** under **Parul University, Vadodara, Gujarat (India)**.

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PIT, Vadodara

DECLARATION

We the undersigned solemnly declare that the project report “**TASKFLOW**” is based on our own work carried out during the course of our study under the supervision of **PRAVESH DWIVEDI, Assistant Professor**
Department of CSE.

We assert the statements made and conclusions drawn are the outcomes of my own work.
I further certify that

1. The work contained in the report is original and has been done by us under the general supervision of our supervisor.
2. The work has not been submitted to any other Institution for any other degree / diploma / certificate in this university or any other University of India or abroad.
3. We have followed the guidelines provided by the university in writing the report.

Whenever we have used materials (data, theoretical analysis, and text) from other sources, we have given due credit to them in the text of the report and giving their details in the references.

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ACKNOWLEDGEMENT

In this semester, we have completed our project on “**TASKFLOW**”. During this time, all the group members collaboratively worked on the project and learnt about the industry standards that how projects are being developed in IT Companies. We also understood the importance of teamwork while creating a project and got to learn the new technologies on which we are going to work in near future.

We gratefully acknowledge for the assistance, cooperation guidance and clarification provided by “**PRAVESH DWIVEDI**” during the development of our project. We would also like to thank our Head of Department **Prof. Sumitra Menaria** and our Principal **Dr. Swapnil Parikh** Sir for giving us an opportunity to develop this project. Their continuous motivation and guidance helped us overcome the different obstacles for completing the Project.

We perceive this as an opportunity and a big milestone in our career development. We will strive to use gained skills and knowledge in our best possible way and we will work to improve them.

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LIST OF ABBREVIATIONS

S. No.	Abbreviation	Full Form
1	AI	Artificial Intelligence
2	API	Application Programming Interface
3	JWT	JSON Web Token
4	REST	Representational State Transfer
5	CRUD	Create, Read, Update, Delete
6	ER	Entity-Relationship
7	DFD	Data Flow Diagram
8	UI	User Interface
9	UX	User Experience
10	MERN	MongoDB, Express.js, React, Node.js
11	JSON	JavaScript Object Notation
12	HTTP	HyperText Transfer Protocol
13	URL	Uniform Resource Locator
14	DB	Database

ABSTRACT

TaskFlow is a modern educational management system designed to streamline interaction between students, faculty, and administrators through a centralized digital platform. The system is developed using the MERN stack (MongoDB, Express.js, React, and Node.js) and focuses on improving task management, attendance tracking, and overall academic workflow.

The platform follows a role-based architecture consisting of three primary users: Admin, Faculty, and Student. The Admin plays a crucial role in managing and maintaining the system by approving new account registrations, creating temporary accounts for faculty members, approving profile changes, and broadcasting important announcements to both students and faculty. Faculty members can create and manage tasks, review student submissions, and monitor performance, while students can view tasks, submit their work, and track their academic progress.

A key feature of the system is the implementation of a geo-fenced attendance system, which ensures that attendance is marked only when students are physically present within a defined location range. This enhances the authenticity and reliability of attendance records.

The system also includes an adaptive learning module powered by Google Generative AI, which is currently under development. This module is intended to provide personalized learning assistance and intelligent explanations based on student performance and learning patterns.

The backend is built using Node.js and Express, with MongoDB used for efficient data management. Secure authentication is implemented using JSON Web Tokens (JWT) and encrypted password storage. The frontend is developed using React, offering a responsive and user-friendly interface.

Overall, TaskFlow provides a scalable, secure, and efficient solution for academic management by integrating modern technologies and intelligent system design.

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CHAPTER I: INTRODUCTION

1.1 Overview

In the modern education system, managing academic activities efficiently has become a significant challenge for institutions. Traditional methods of handling tasks, attendance, and student performance often involve manual processes, which are time-consuming, error-prone, and lack transparency. With the rapid growth of digital technologies, there is a need for smart systems that can streamline communication and automate academic workflows.

TaskFlow is designed as a centralized educational management platform that connects students, faculty, and administrators in a unified system. The platform simplifies the process of task assignment, submission, attendance tracking, and performance monitoring. It is developed using the MERN stack (MongoDB, Express.js, React, and Node.js), ensuring scalability, flexibility, and efficient data handling.

The system follows a role-based architecture with three main users: Admin, Faculty, and Student. Each role has specific functionalities. The Admin manages user accounts, approves registrations, creates temporary accounts for faculty, and broadcasts announcements. Faculty members can assign tasks, review submissions, and monitor student progress. Students can access tasks, submit their work, and track their performance.

A key highlight of TaskFlow is the geo-fenced attendance system, which ensures that students can only mark attendance when they are physically present within a specific location. This enhances the reliability and authenticity of attendance records.

Additionally, the system includes an adaptive learning module powered by AI, which is currently under development and aims to provide personalized learning support.

Overall, TaskFlow provides an efficient, secure, and scalable solution for academic management by integrating modern technologies and intelligent system design.

1.2 Problem Statement

Existing academic management systems face several challenges that reduce their effectiveness and usability:

- **Manual Processes:** Many institutions still rely on manual methods for attendance and task management, leading to inefficiencies and human errors.
- **Lack of Centralization:** Academic activities such as assignments, attendance, and communication are often handled through multiple platforms, creating confusion and inconsistency.
- **Unreliable Attendance Systems:** Traditional attendance systems can be easily manipulated and do not ensure the physical presence of students.
- **Poor Communication:** There is no unified system for broadcasting important information to all students and faculty members.

- **Limited Administrative Control:** Existing systems often lack proper control mechanisms for managing user accounts and monitoring activities.
- **Lack of Personalization:** Most platforms do not provide intelligent or adaptive learning support based on student performance.

These issues highlight the need for a comprehensive system that integrates all academic activities into a single, efficient platform.

1.3 Objective of Project

The main objectives of TaskFlow are:

1. To develop a centralized platform for managing academic tasks, attendance, and communication.
2. To implement a secure authentication system using JWT for safe user access.
3. To design a role-based system with Admin, Faculty, and Student functionalities.
4. To enable faculty members to create, manage, and evaluate student tasks efficiently.
5. To allow students to submit assignments and track their academic progress.
6. To implement a geo-fenced attendance system to ensure accurate and location-based attendance marking.
7. To provide administrative control for approving user registrations, managing accounts, and broadcasting announcements.
8. To integrate an AI-based adaptive learning module for personalized learning support (under development).
9. To build a responsive and user-friendly interface for better user experience.

1.4 Applications and Scope

TaskFlow has a wide range of applications in the field of education and academic management.

Applications:

- Educational institutions for managing student and faculty activities
- Colleges and universities for task and attendance management
- Training institutes for assignment tracking and performance monitoring
- Online learning platforms for structured workflow management

Scope:

- The system currently supports web-based usage with role-based access control.
- It is designed to handle multiple users simultaneously with secure data management.
- The geo-fenced attendance feature ensures real-world applicability in institutions.

- The AI module provides scope for future enhancement in personalized education.
- The platform can be further extended into a mobile application and deployed on cloud infrastructure.

1.5 Organization of Report

This report is organized into multiple chapters to provide a clear understanding of the project:

Chapter I (Introduction): Provides an overview of the project, problem statement, objectives, and scope.

Chapter II (Literature Survey): Discusses existing systems, related technologies, and research work.

Chapter III (Methodology): Explains system architecture, technologies used, modules, and diagrams.

Chapter IV (System Requirements): Describes hardware and software requirements for the system.

Chapter V (Expected Outcomes): Presents system features, performance expectations, and interface details.

Chapter VI (Conclusion and Future Scope): Summarizes the project and discusses possible future enhancements.

Chapter VII (References): Lists all sources and materials referred to during the project.

Chapter VIII (Weekly Report): Provides a timeline of project development and progress.

CHAPTER II: LITERATURE SURVEY

The rapid advancement of technology has significantly transformed the education sector. Digital platforms are increasingly being used to manage academic activities such as attendance tracking, assignment submission, and communication between students and faculty. This chapter reviews existing systems, technologies, and research related to educational management systems and identifies gaps that led to the development of TaskFlow.

Existing Systems:

Several platforms currently exist to support academic and learning management:

• Google Classroom:

A widely used platform that allows teachers to create and manage assignments, share resources, and communicate with students. However, it lacks advanced administrative control and location-based attendance features.

• Moodle:

An open-source learning management system that provides course management, quizzes, and assignments. While powerful, it can be complex to use and lacks modern UI and intelligent automation features.

• Microsoft Teams (Education):

Provides communication, collaboration, and assignment features. However, it is primarily focused on communication and lacks dedicated academic workflow optimization and attendance validation.

• Traditional ERP Systems:

Used by many institutions for attendance and academic management, but they often rely on manual data entry and lack real-time validation and modern user experience.

Limitations of Existing Systems:

Despite the availability of these systems, several limitations still exist:

- Lack of a unified platform that integrates tasks, attendance, and communication
- Absence of reliable attendance systems that ensure physical presence
- Limited administrative control over user activities and approvals
- Poor user experience in traditional systems
- Lack of intelligent or adaptive learning support
- Dependency on multiple tools instead of a single centralized solution

Technologies Used in Existing Solutions:

Modern academic systems are built using various technologies:

- **Web Technologies:** HTML, CSS, JavaScript for frontend development
- **Frameworks:** React, Angular for dynamic and responsive interfaces
- **Backend Technologies:** Node.js, Django, Spring Boot
- **Databases:** MySQL, PostgreSQL, MongoDB
- **Authentication:** JWT and OAuth for secure access
- **Cloud Services:** AWS, Firebase for deployment and scalability

These technologies enable scalability, security, and efficient performance of web-based systems.

Research and Trends:

Recent trends in educational technology include:

- **AI-based Learning Systems:** Personalized learning platforms using artificial intelligence
- **Automation in Education:** Reducing manual work through digital workflows
- **Cloud-based Platforms:** Accessible systems from anywhere
- **Data Analytics:** Monitoring student performance and generating insights
- **Location-based Services:** Use of geolocation for attendance tracking

These trends highlight the importance of integrating intelligent and automated features into academic systems.

Motivation for Proposed System :

The analysis of existing systems and technologies reveals a gap in providing a complete, efficient, and intelligent academic management solution. Most platforms focus on limited functionalities and fail to integrate all essential features into a single system.

TaskFlow is proposed to overcome these limitations by:

- Providing a centralized platform for all academic activities
- Implementing a geo-fenced attendance system for accurate tracking
- Introducing an admin-controlled environment for better system management
- Enabling task management and performance tracking in a structured way
- Integrating AI-based learning support (under development)

CHAPTER III: METHODOLOGY

3.1 Background and Overview of Methodology

The development of TaskFlow follows a structured approach combining modern web development practices with a modular system design. The methodology focuses on building a scalable, secure, and efficient platform that integrates multiple academic functionalities into a single system.

The project is developed using a full-stack approach where the frontend, backend, and database are interconnected through APIs. The system follows a client-server architecture, where the frontend handles user interaction, the backend processes requests and business logic, and the database stores and retrieves data.

The development process includes:

- **Requirement Analysis:** Understanding user needs such as task management, attendance tracking, and admin control
- **System Design:** Designing architecture, database schema, and workflows
- **Implementation:** Developing frontend and backend modules
- **Testing:** Verifying functionality and fixing errors
- **Integration:** Connecting all modules into a working system

This methodology ensures smooth development and proper organization of system components.

3.2 Project Platforms and Technologies Used

Component	Technology
Frontend Framework	React.js
Styling	Tailwind CSS
Backend Framework	Node.js + Express.js
Database	MongoDB
Authentication	JWT (JSON Web Tokens)
Password Security	BcryptJS
File Handling	Multer
API Communication	Axios

3.3 Proposed Methodology:

TaskFlow follows a role-based and modular approach:

1. **User Authentication:**
Users register and log in securely using JWT-based authentication.
2. **Role-Based Access Control:**
The system differentiates between Admin, Faculty, and Student roles, providing specific functionalities to each.
3. **Task Management:**
Faculty creates tasks, and students submit their work through the system.
4. **Attendance System:**
Geo-fencing is used to verify the physical presence of students before marking attendance.

5. **Admin Control System:**

Admin manages user approvals, creates temporary faculty accounts, and broadcasts announcements.

6. **Adaptive Learning Module:**

AI-based learning assistance is integrated but currently under development.

This structured approach ensures efficient workflow and proper system management.

3.4 Project Modules

The TaskFlow system is divided into several modules:

1. **Authentication Module:**

Handles user registration, login, and session management.

2. **Admin Module:**

- Approves new user registrations
- Creates temporary accounts for faculty
- Approves profile updates
- Sends broadcast messages to all users

3. **Faculty Module:**

- Creates and manages tasks
- Reviews student submissions
- Monitors performance

4. **Student Module:**

- Views assigned tasks
- Submits assignments
- Tracks progress

5. **Attendance Module:**

- Uses geolocation to verify student presence
- Records attendance data securely

6. **Adaptive Learning Module (Under Development):**

- Intended to provide AI-based learning support

3.5 System Architecture Diagrams

System Architecture:

The system follows a three-tier architecture:

- Presentation Layer: React-based frontend
- Application Layer: Node.js and Express backend
- Data Layer: MongoDB database

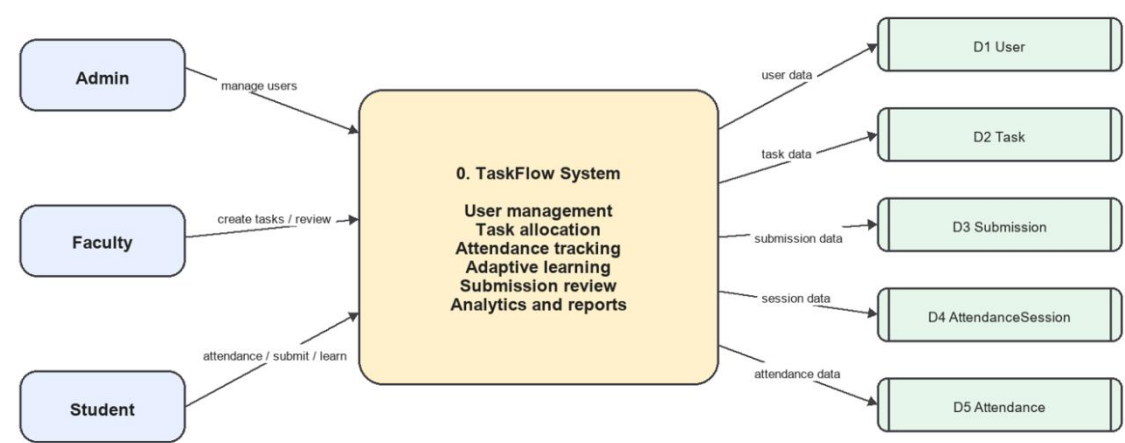
Data Flow:

User → Frontend → Backend (API) → Database

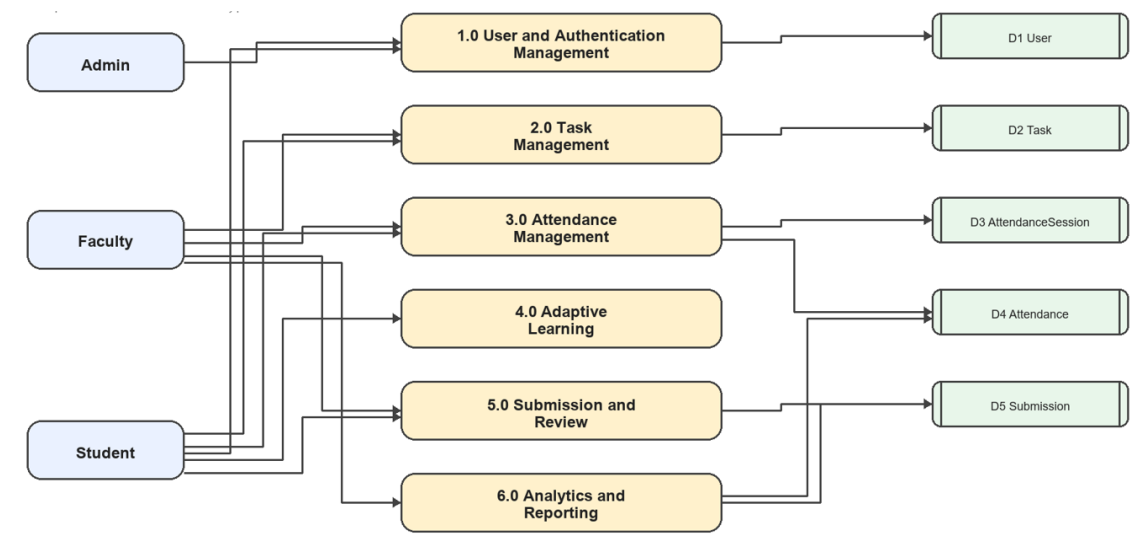
Database → Backend → Frontend → User

Additionally, the system integrates external AI services for future adaptive learning features.

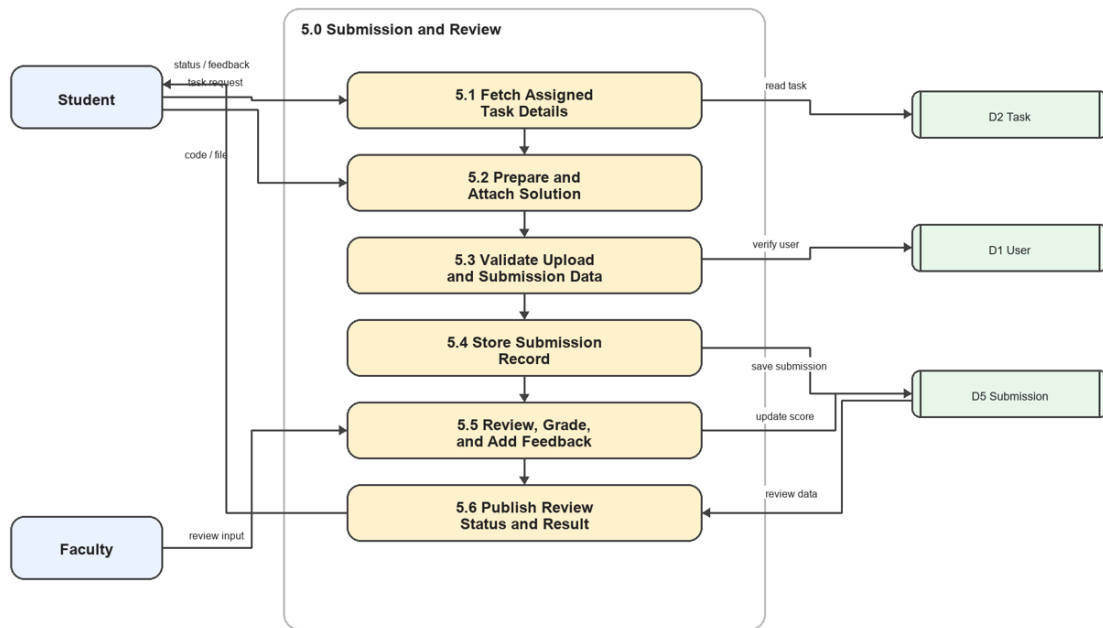
Level 0:



Level 1:



Level 2:



Entity Relationships:

Entities:

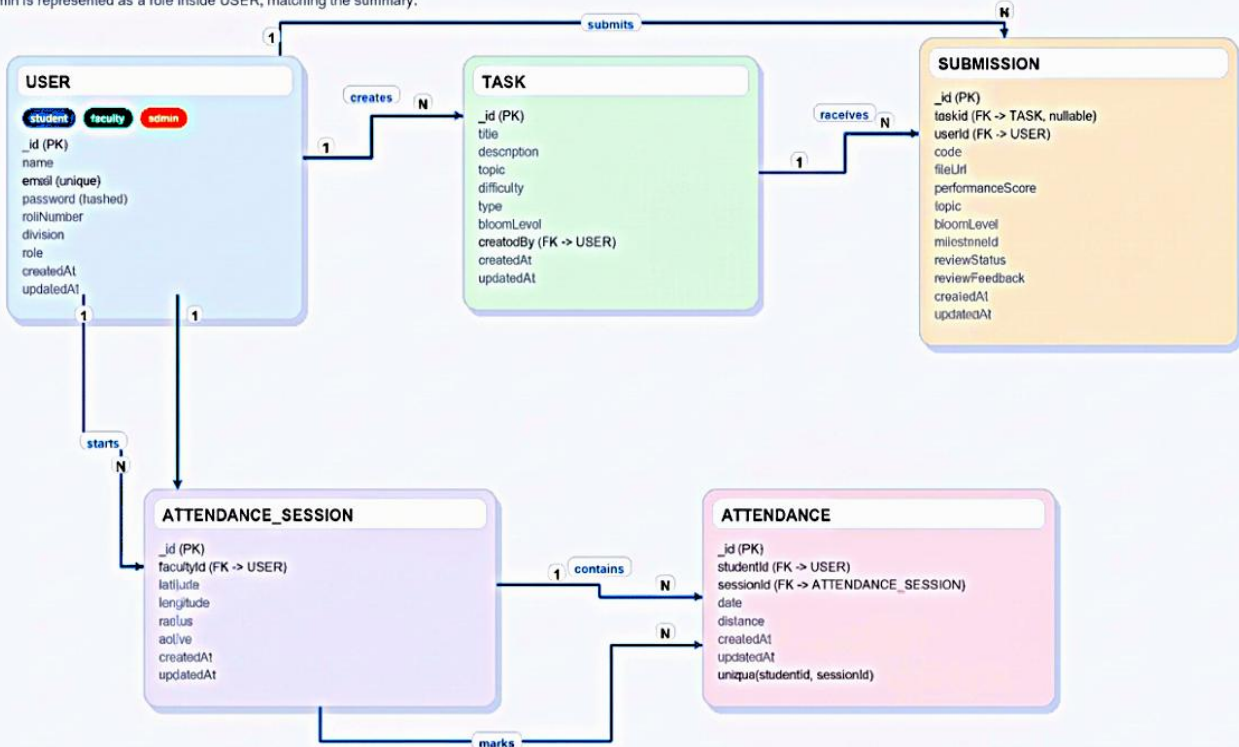
- User
- Task
- Submission
- AttendanceSession
- Attendance

Relations:

- User → Submission (1:N)
- Task → Submission (1:N)
- Session → Attendance (1:N)

TaskFlow Entity Relationship Diagram

Redrawn from the project summary in ERD form with clear non-overlapping relationship lines.
Admin is represented as a role inside USER, matching the summary.



API Endpoints Summary:

Module	Method	Endpoint	Description
Auth	POST	/api/auth/register	Register new user
Auth	POST	/api/auth/login	User login
Auth	GET	/api/auth/profile	Get user profile

Admin	POST	/api/admin/approve-user	Approve user registration
Admin	POST	/api/admin/create-faculty	Create temporary faculty account
Admin	POST	/api/admin/broadcast	Send announcement

Task	POST	/api/tasks/create	Create new task
Task	GET	/api/tasks	Get all tasks
Task	POST	/api/tasks/submit	Submit task

| Submission | GET | /api/submissions | View submissions |
| Submission | POST | /api/submissions/review | Review submission |

| Attendance | POST | /api/attendance/mark | Mark attendance |
| Attendance | GET | /api/attendance/session | Get attendance session |

CHAPTER IV: SYSTEM REQUIREMENTS

4.1 Software Requirements

Software	Version/Details
Operating System	Windows/macOs/Linux
Frontend Network	React.js (18.x)
Styling	Tailwind CSS
Backend	Node.js(v18 or higher)
Backend Framework	Express.js
Database	MongoDB(Local/Atlas)
API Testing Tool	Postman
Version Control	Git & GitHub
Code Editor	Visual Studio Code
Browser	Google Chrome/Firefox/Safari
Authentication	JWT(JSON Web Token)
Package Manager	npm

4.2 Hardware Requirements

Component	Specification(Recommended)
Processor	Intel i5/Ryzen 5 or equivalent (2 GHz+)
RAM	8 GB minimum (16 GB recommended)
Storage	10 GB SSD for development environment
Network	Stable Internet connection (2+ Mbps)

CHAPTER V: EXPECTED OUTCOMES

5.1 Key Features and Deliverables

The TaskFlow system is expected to provide the following features and functionalities:

1. **Role-Based System:**
 - Three types of users: Admin, Faculty, and Student
 - Secure login and authentication system
2. **Admin Functionalities:**
 - Approve new user registrations
 - Create temporary accounts for faculty
 - Approve profile changes
 - Broadcast announcements to students and faculty
3. **Task Management System:**
 - Faculty can create and assign tasks
 - Students can view and submit assignments
 - Faculty can review and evaluate submissions
4. **Attendance System:**
 - Geo-fenced attendance tracking
 - Ensures students are physically present
 - Secure and reliable attendance records
5. **User Dashboard:**
 - Separate dashboards for Admin, Faculty, and Student
 - Overview of tasks, submissions, and activities
6. **Adaptive Learning Module (Under Development):**
 - AI-based learning assistance
 - Personalized recommendations and explanations

5.2 Expected System Performance

The TaskFlow system is designed to deliver efficient and reliable performance:

- Fast response time for login and data retrieval
- Smooth interaction between frontend and backend
- Secure handling of user data using JWT authentication
- Efficient database operations using MongoDB
- Ability to handle multiple users simultaneously

Expected Performance Metrics:

- Login Response Time: Less than 2 seconds
- Data Fetch Time: Less than 1 second
- API Response Time: 200–500 ms (approx.)
- System Availability: High reliability under normal usage

5.3 Expected User Interfaces

1. Login / Signup Page:

- Secure authentication system
- User-friendly input forms

LOGIN

SIGN UP

Sign Up

Student

Studentdemo

demostudent@taskflow.com

Password@123

Strong — Use 8+ chars with uppercase, lowercase & numbers

Create Account

2. Admin Dashboard:

- User approval system

TASKFLOW

ADMINISTRATION

Dashboard

Students Overview

Profile Approvals

Account Approvals

Announcements

Settings

Logout

Admin Portal

System Admin Admin

Institution Overview

Management dashboard for system-wide monitoring and controls.

New Announcement

TOTAL

1

Total Students

+1 this month

ACTIVE

1

Active Faculty

Full staff active

ACCOUNT

0

Account Requests

Action required

PROFILE

2

Profile Requests

Verification needed

ANNOUNCEMENTS

2

Announcements

System wide

Account Requests

0 Pending

- Broadcast management
- Account management

TASKFLOW

MAIN

Dashboard

Students

Tasks

Reviews

ACADEMICS

Attendance

Performance Report

Logout

Manage Tasks

Create New Task

Publish Task

Topic

e.g. Data Structures

Type

Task

Difficulty

Medium

Bloom Level

Remember

Tip: Provide clear instructions and formatting to help students better understand the task requirements.

System Admin

Admin

Task Title...

Normal | | **B** | *I* | U | | | | | |

Write the task details, attachments, and instructions here...

TASKFLOW

MAIN

Dashboard

Students

Tasks

Reviews

ACADEMICS

Attendance

Performance Report

Log out

Faculty Dashboard

System Admin
Admin

Upload Study Material

Manage upload study material

Open

Quick Upload

Student Overview

View class-wise student insights

Open

Mark Attendance

Manage mark attendance

Open

Create / Evaluate Test

Manage create / evaluate test

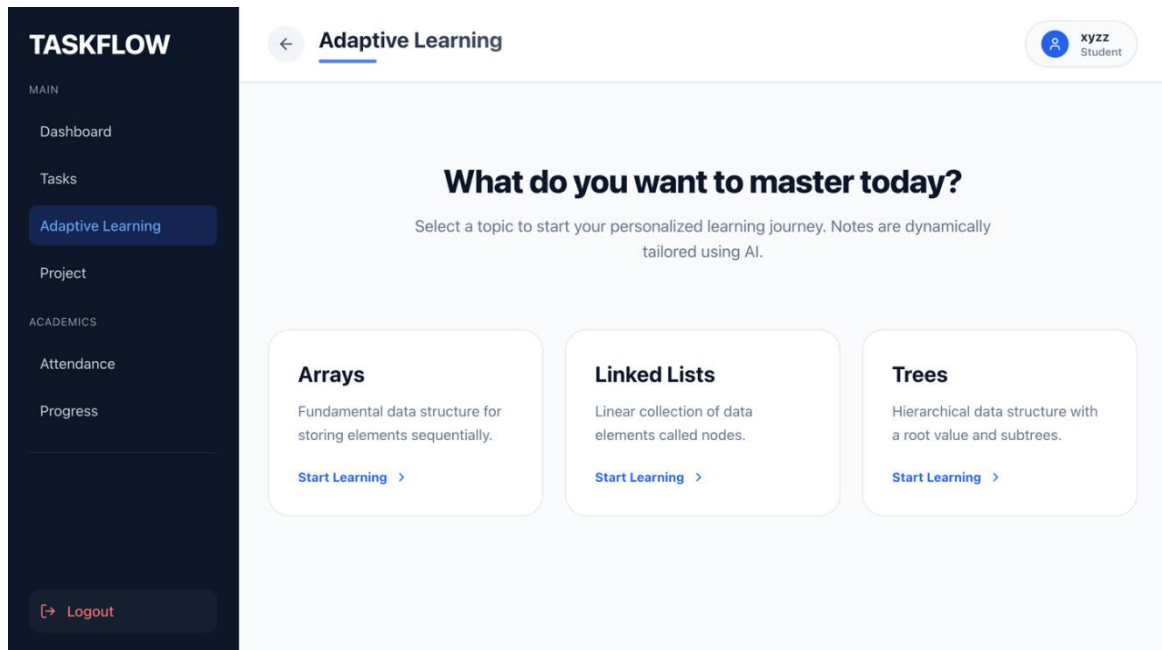
Open

Project Reviews

Manage project reviews

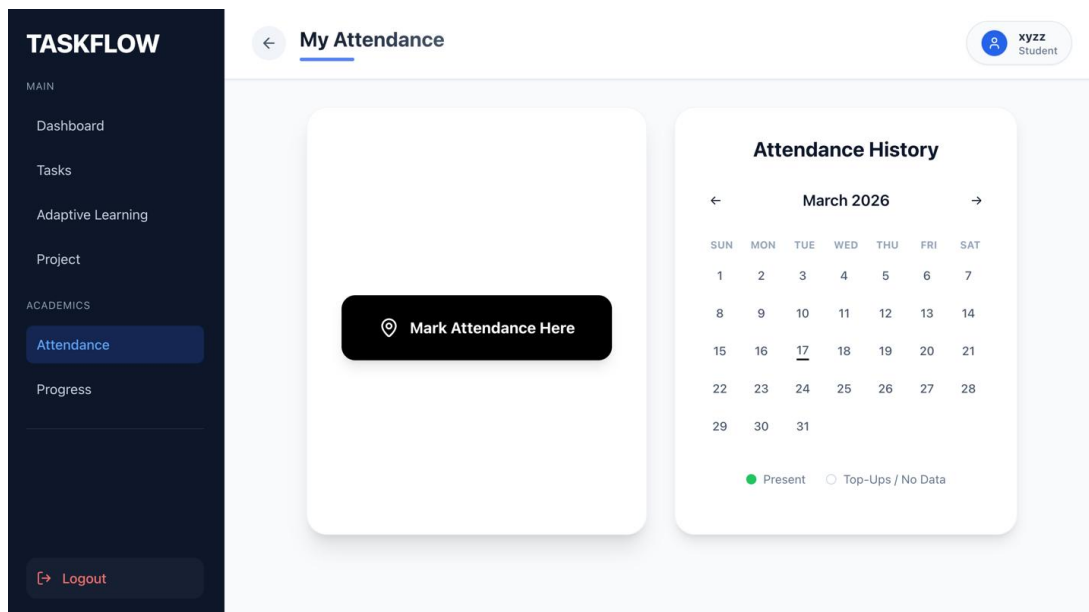
Open

Create New Task



5. Attendance Interface:

- Location-based attendance marking
- Real-time validation of user location



Task Creation Interface

- Allows faculty members to create new tasks
- Enables assignment of tasks to students
- Includes fields for task details such as topic and difficulty level

- Helps in organizing and managing academic activities efficiently

TASKFLOW

MAIN

- Dashboard
- Students
- Tasks**
- Reviews

ACADEMICS

- Attendance
- Performance Report

[Logout](#)

← **Manage Tasks**

System Admin Admin

← **Create New Task** [Publish Task](#)

Topic: e.g. Data Structures

Type: Task

Difficulty: Medium

Bloom Level: Remember

Tip: Provide clear instructions and formatting to help students better understand the task requirements.

Task Title...

Normal B I U G " | | | | |

Write the task details, attachments, and instructions here...

5.4 Expected Outcomes

The TaskFlow system is expected to achieve the following outcomes:

- Simplification of academic management processes
- Improved communication between students, faculty, and admin
- Increased accuracy and reliability of attendance records
- Efficient task management and evaluation system
- Enhanced user experience through a centralized platform

CHAPTER VI: CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

TaskFlow is a comprehensive academic management system designed to simplify and enhance interaction between students, faculty, and administrators. The project successfully implements a centralized platform that integrates task management, attendance tracking, and administrative control into a single system.

The system demonstrates the effective use of modern web technologies such as the MERN stack, enabling a responsive and scalable application. Key features such as role-based access control, secure authentication using JWT, and structured task workflows contribute to efficient academic operations.

One of the major achievements of the project is the implementation of a geo-fenced attendance system, which ensures that attendance is recorded only when students are physically present within a defined location. This significantly improves the reliability and authenticity of attendance data.

The inclusion of an Admin module adds strong control over the system by allowing user approval, account management, and communication through broadcast messages. This makes the platform more organized and suitable for real-world institutional use.

Although the adaptive learning module powered by AI is currently under development, its integration reflects the future direction of the system towards intelligent and personalized learning.

Overall, TaskFlow successfully achieves its objective of providing a secure, efficient, and user-friendly solution for academic management.

6.2 Future Scope

The TaskFlow system has significant potential for further enhancement and expansion. Some of the future improvements include:

- **AI-Based Adaptive Learning:**

Complete development of the AI module to provide personalized learning recommendations and intelligent feedback.

- **Mobile Application Development:**

Development of Android and iOS applications for improved accessibility and user convenience.

- **Cloud Deployment:**

Deploying the system on cloud platforms such as AWS or Azure for better scalability and availability.

- **Real-Time Notifications:**

Integration of real-time notifications using technologies like WebSockets for instant updates.

- **Advanced Analytics Dashboard:**

Implementation of detailed analytics for tracking student performance and engagement.

- **Integration with External Systems:**

Connecting with LMS platforms, email services, or academic databases for extended functionality.

- **Enhanced Security Features:**

Adding multi-factor authentication and advanced encryption techniques.

- **Multi-Language Support:**

Providing support for regional languages to make the system more accessible.

These future enhancements will make TaskFlow more powerful, scalable, and suitable for large-scale deployment in educational institutions.

Final Remarks

TaskFlow represents a step towards modernizing academic management systems by combining automation, security, and intelligent features. With further development and enhancements, it has the potential to evolve into a full-scale educational platform.

CHAPTER VII: REFERENCES

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- React Official Documentation – <https://react.dev/>
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- Express.js Documentation – <https://expressjs.com/>
- MDN Web Docs – <https://developer.mozilla.org/>
- JSON Web Token (JWT) – <https://jwt.io/introduction>
- Mongoose Documentation – <https://mongoosejs.com/docs/>
- Tailwind CSS Documentation – <https://tailwindcss.com/docs>
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- Google Generative AI Documentation – <https://ai.google.dev/>
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- BcryptJS Documentation – <https://www.npmjs.com/package/bcryptjs>

CHAPTER VIII: WEEKLY REPORT

Weekly Report: Project Development Progress

Week 1-2: Requirement Analysis and Planning

- Identified project objectives and system requirements
 - Discussed features such as task management, attendance, and admin control
 - Designed initial project structure
-

Week 3-4: System Design

- Designed system architecture and database schema
 - Planned frontend and backend structure
 - Finalized technology stack (MERN)
-

Week 5-6: Frontend Development

- Developed login and registration pages
 - Created dashboards for Admin, Faculty, and Student
 - Implemented UI using React and Tailwind CSS
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Week 7-8: Backend Development

- Set up Node.js and Express server
 - Implemented API routes and controllers
 - Integrated MongoDB database using Mongoose
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Week 9-10: Core Features Implementation

- Developed task management system
 - Implemented submission and review features
 - Built geo-fenced attendance system
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Week 11: Integration and Testing

- Connected frontend with backend APIs
 - Tested all modules and fixed bugs
 - Verified authentication and data flow
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Week 12: Documentation and Finalization

- Prepared project report and documentation